

Airbnb Bookings Predictions

SVM and Neural Network Model Performance Analysis

Pamela Afful

Problem Statement

A new competitor in the private residential market is interested in the Airbnb market in Cape Town and would like an analysis on the key predictors of demand of/ availability of Airbnbs. Demand in this case is proxied by the target variable, `fully_booked_30`, which is a binary variable that populates 1 if the Airbnb is fully booked in the next 30 days and 0 if not. This is done by using and comparing the classification metrics across 3 classification models namely:

- SVM with a linear kernel
- SVM with a radial kernel
- Feed-forward Neural Network

The insight will serve as a basis to gauge if there is viable demand for the competitor to enter the market.

10-fold cross-validation was used to train all the models, each with its own set of hyperparameters. The summary of the best-performing model for each is then selected and presented, along with a plot of the average CV metric across hyperparameter combinations.

Exploratory Analysis

The data is highly imbalanced in favour of the negative class. Since the modelling aims to predict availability, this will be treated as the positive class in modelling- this is the target class.

The data used was listings data containing key attributes across Airbnb IDs.

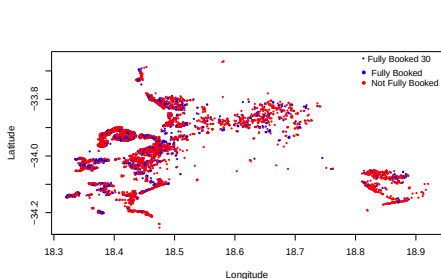


Figure 1: Spatial Distribution of Listings Grouped by Target

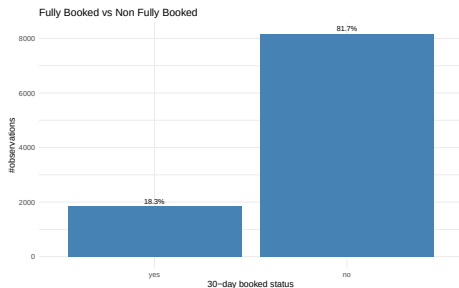


Figure 2: Distribution of Target, Fully_booked_30

Exploratory Analysis Feature Distributions

When grouping by the target variable, `fully_booked_30`, the following key insights are observed:

- Average Price and average minimum nights are slightly higher for observations in the “yes” class, with the gap much wider for average minimum nights (~ 50 % higher than the “no” class).
- Average review scores and average response rate are fairly evenly split across the two target classes
- Average host listings are higher for the “no” class with an average of ~20 listings.
- For the host age, the median (118 months) host age is ~5% higher for the “yes” class compared to the no class. Moreover no class has a longer range of host ages, although the distributions about the median for both groups are similar; most host ages lie below the median host age.

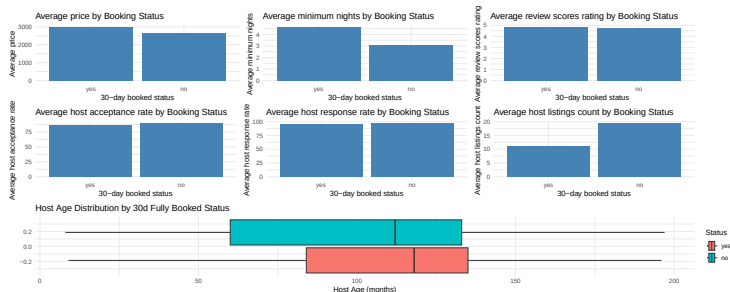


Figure 3: Feature Averages and Distributions Grouped by Target

Hyperparameter definitions

Cost Parameter: This is a constant >0 controlling how strict (alternatively, how wide or narrow) the margin is around the linear decision boundary. A higher cost value indicates a stricter model (and thus a narrower margin), whilst a low cost indicating a more tolerant model (wider margin). For the linear kernel SVM, the cost parameter is tuned with the following options:

1. Sequence starting from 0.15 to 3, in steps of 0.15, amounting to 20 cost parameter options.

Table 1: Linear Kernel SVM Best Model Summary

Metric	Value
best_cost	0.15
nr.support_vectors	3807
best_cross_validation_error	0.18
best_training_accuracy	0.82

Model Training Linear Kernel SVM

Linear Kernel SVM Gridsearch Summary

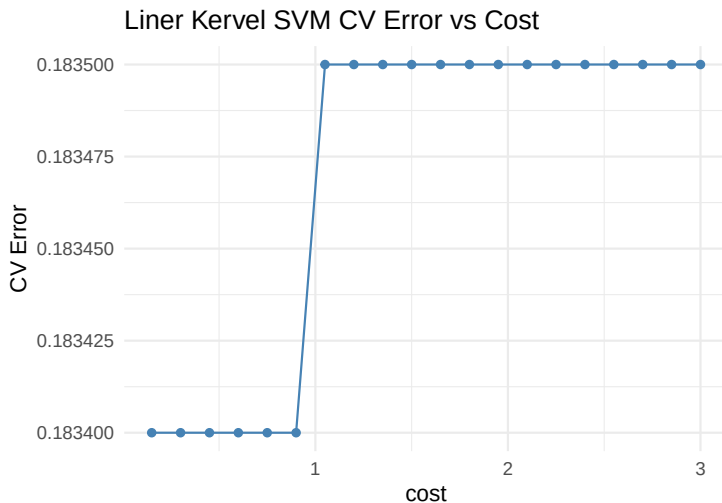


Figure 4: Linear Kernal SVM CV Error vs Cost

A radial-kernel SVM was chosen to train on the data. This kernel has an additional similarity parameter which controls the rate of similarity between points based on their distance; thus, this needs to be tuned during the cross-validation process.

Table 2: Radial Kernel SVM Best Model summary

Metric	Value
nr.support_vectors	7421.00
best_cost	0.75
best_gamma	0.45
best_cross_validation_error	0.18
best_training_accuracy	0.85

Model Training Non-Linear Kernel SVM

Gridsearch Training Summary

Across parameter combinations, the model starts to overfit as the gamma parameter increases past 0.50, as well as some higher values of gamma, past its optimal value.

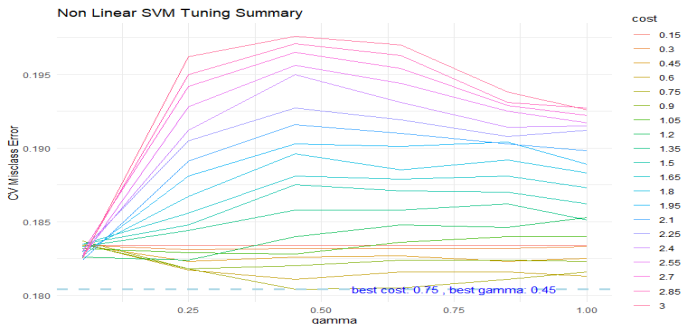


Figure 5: Non-linear Kernel SVM CV Error Per Grid Hyperparameter

```
(nl_svm_model$fitted)
```

NULL

```
nl_svm_model$best.perform
```

```
[1] 0.1804
```

Feed Forward Neural Network

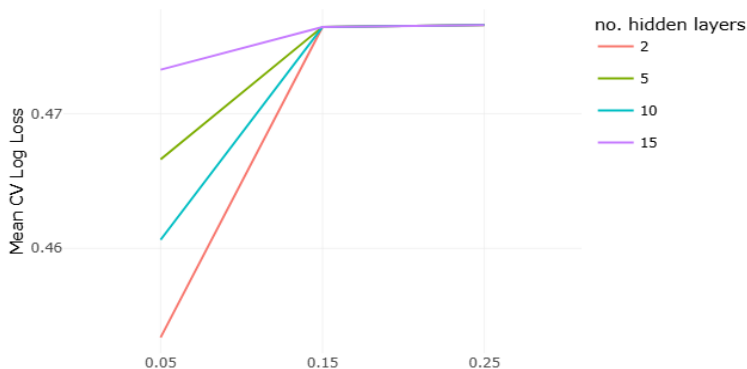
Grid Search Tuning Summary

Grid search was done on the following hyperparameters:

- 1 Number of hidden layer nodes: (2, 5, 10, 15)
- 2 L1: (0.5, 0.15, 0.25)
- 3 Number of Training Epochs: (500, 1000, 1500, 2000)

The candidate model with the lowest CV log loss is chosen.

NN Gridsearch Summary



Best Model Summary

Table 3: Neural Net Best Model Metrics

Metric	Value
best L1	0.05
best nr. hidden layer nodes	2.00
best CV Accuracy	0.82
best CV Logloss	0.45

Key Considerations

Since the data is imbalanced and we are mostly interested in predicting the majority class (fully booked=no), specificity with the main metric used to distinguish the best performing model***look this up!!

Table 4: Cross Model CV Metrics

Model	Specificity	Sensitivity	F1	Bal.Accuracy
Linear Kernal SVM	0.00	1.00	0.90	0.50
Radial Kernel SVM	0.20	1.00	0.92	0.60
Feed Forward NN	0.73	0.51	0.65	0.62

